

NORTHER UNIVERSITY BANGLADESH
Department of Computer Science & Engineering

LAB PROJECT REPORT

CSE-2367 (ECSE)

Topic: NUB Club & Event Management System

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Submitted To

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Introduction & Background

At Northern University Bangladesh (NUB), student clubs play a significant role in promoting academic excellence, cultural engagement, leadership development, and technical innovation. These clubs regularly organize various activities, including seminars, workshops, competitions, cultural programs, and training sessions. However, managing club memberships, event schedules, participant registrations, and related records through manual processes often leads to data inconsistency, redundancy, and increased administrative workload.

To overcome these challenges, the **NUB Club & Event Management System** has been designed as a comprehensive relational database management solution. The system integrates student information, club records, membership details, event management, and registration processes into a centralized and automated platform. By utilizing structured database techniques, the system ensures efficient data organization, improved accuracy, secure record maintenance, and streamlined administrative operations.

The proposed system not only minimizes manual effort but also enhances data integrity and supports effective decision-making through reliable data retrieval and reporting functionalities.

Project Objectives

The primary objectives of the **NUB Club & Event Management System** are:

- To store, organize, and manage detailed student information along with their associated club memberships.
- To maintain a structured database of clubs, including club organizers, executive panels, and member roles.
- To manage club events efficiently while enforcing event capacity limitations and scheduling constraints.
- To record student event registrations accurately and prevent duplicate registration entries.
- To facilitate advanced querying, reporting, and analytical operations across the entire management system.
- To improve administrative efficiency, reduce data redundancy, and ensure consistency within the database environment.
- To provide a scalable and reliable database framework that can support future enhancements and additional university club activities.

System Entities:

Entity	Attributes
Student	StudentID, Name, Email, Phone
Club	ClubID, ClubName, Admin
Membership	StudentID (FK), ClubID (FK), Role (admin/member)
Event	EventID, Title, OrganizingClubID (FK), EventDate, Capacity
Registration	RegistrationID, StudentID (FK), EventID (FK), RegistrationDate, Attended

E-R Diagram :

An Entity-Relationship diagram has been constructed to represent the overall structure of the database. The diagram was created in Draw.io and its key components are outlined below:

Entities and Relationships :

Entities:

- STUDENT (StudentID [PK], Name, Email, Phone)
- CLUB (ClubID [PK], ClubName, Admin)
- EVENT (EventID [PK], Title, EventDate, Capacity, OrganizingClubID [FK])

Relationships:

- STUDENT — MEMBERSHIP — CLUB (Many-to-Many with Role attribute)
- CLUB — organizes — EVENT (One-to-Many)
- STUDENT — REGISTRATION — EVENT (Many-to-Many with RegistrationDate, Attended attributes)

Relational Schema:

Based on the E-R diagram developed earlier, the relational schema below was constructed to define the structure of each database table:

Tables

Student :

Student(StudentID [PK], Name, Email, Phone)

Club

Club(ClubID [PK], ClubName, Admin)

Membership:

Membership(StudentID [PK, FK → Student], ClubID [PK, FK → Club], Role) The combination of StudentID and ClubID forms a composite primary key, which prevents any student from being enrolled in the same club more than once.

Event :

Event(EventID [PK], Title, OrganizingClubID [FK → Club], EventDate, Capacity)

Registration:

Registration(RegistrationID [PK], StudentID [FK → Student], EventID [FK → Event], RegistrationDate, Attended) UNIQUE(StudentID, EventID) -- prevents duplicate registration

Foreign Key Relationships:

Table	Foreign Key	References
Membership	StudentID	Student(StudentID)
Membership	ClubID	Club(ClubID)
Event	OrganizingClubID	Club(ClubID)
Registration	StudentID	Student(StudentID)
Registration	EventID	Event(EventID)

Normalization (Up to 3NF):

First Normal Form (1NF)

The database schema satisfies the requirements of **First Normal Form (1NF)** based on the following conditions:

- All attributes contain atomic values, meaning that no multi-valued or composite attributes exist within any table.
 - Each table is defined with a unique and clearly identifiable primary key.
 - The schema does not include repeating groups, nested records, or array-type attributes.
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Second Normal Form (2NF)

The schema is also compliant with **Second Normal Form (2NF)** for the following reasons:

- Tables such as **Student**, **Club**, **Event**, and **Registration** use single-attribute primary keys; therefore, they automatically satisfy the requirements of 2NF.
 - The **Membership** table uses a composite primary key consisting of (**StudentID**, **ClubID**). The non-key attribute **Role** is fully functionally dependent on the entire composite key rather than on any individual part of it. As a result, partial dependency is eliminated, ensuring compliance with 2NF.
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Third Normal Form (3NF)

All relations within the database conform to **Third Normal Form (3NF)** because no transitive dependencies are present among non-key attributes.

- In the **Student** table, the attributes **Name**, **Email**, and **Phone** depend directly and exclusively on **StudentID**.
- In the **Club** table, both **ClubName** and **Admin** are fully dependent on **ClubID**, with no intermediate dependency.
- In the **Membership** table, the attribute **Role** depends entirely on the composite key (**StudentID**, **ClubID**) without any transitive relationship.
- In the **Event** table, the attributes **Title**, **EventDate**, and **Capacity** are directly determined by **EventID**. The attribute **OrganizingClubID** functions solely as a foreign key and does not create transitive dependency.
- In the **Registration** table, the attributes **RegistrationDate** and **Attended** are directly dependent on **RegistrationID**. Additionally, the **UNIQUE constraint** on (**StudentID**, **EventID**) ensures data consistency while maintaining compliance with 3NF.

SQL Implementation :

Database & Table Creation

Student Table

```
CREATE DATABASE NUB_ClubEvents;
USE NUB_ClubEvents;

CREATE TABLE Student (
  StudentID INT PRIMARY KEY AUTO_INCREMENT,
  Name VARCHAR(100) NOT NULL, Email
  VARCHAR(100) UNIQUE NOT NULL,
  Phone VARCHAR(15)
);
```

Club Table

```
CREATE TABLE Club (
  ClubID INT PRIMARY KEY AUTO_INCREMENT,
  ClubName VARCHAR(100) NOT NULL,
  Admin VARCHAR(100)
);
```

Membership Table

```
CREATE TABLE Membership (
  StudentID INT,
  ClubID INT,
  Role ENUM('admin', 'member') NOT NULL,
  PRIMARY KEY (StudentID, ClubID),
  FOREIGN KEY (StudentID) REFERENCES Student(StudentID),
  FOREIGN KEY (ClubID) REFERENCES Club(ClubID)
);
```

Event Table

```
CREATE TABLE Event (
  EventID INT PRIMARY KEY AUTO_INCREMENT,
  Title VARCHAR(150) NOT NULL,
  OrganizingClubID INT,
  EventDate DATE,
  Capacity INT NOT NULL,
  FOREIGN KEY (OrganizingClubID) REFERENCES Club(ClubID)
);
```

Registration Table

```
CREATE TABLE Registration (
  RegistrationID INT PRIMARY KEY AUTO_INCREMENT,
  StudentID INT,
```

```

EventID          INT,
RegistrationDate  DATE DEFAULT (CURRENT_DATE),
Attended         BOOLEAN DEFAULT FALSE,
UNIQUE          (StudentID, EventID),
FOREIGN KEY (StudentID) REFERENCES Student(StudentID),
FOREIGN KEY (EventID)  REFERENCES Event(EventID)
);

```

Sample Data :

Students

```

INSERT INTO Student (Name, Email, Phone) VALUES
('Alice Rahman', 'alice@nub.ac.bd', '01711000001'),
('Bob Hossain', 'bob@nub.ac.bd', '01711000002'),
('Carol Ahmed', 'carol@nub.ac.bd', '01711000003'),
('David Islam', 'david@nub.ac.bd', '01711000004'),
('Eva Khanam', 'eva@nub.ac.bd', '01711000005'),
('Farhan Uddin', 'farhan@nub.ac.bd', '01711000006');

```

Clubs

```

INSERT INTO Club (ClubName, Admin) VALUES
('Coding Club', 'Alice Rahman'),
('Drama Club', 'Bob Hossain'),
('Robotics Club', 'Carol Ahmed');

```

Memberships

```

INSERT INTO Membership (StudentID, ClubID, Role) VALUES
(1,1,'admin'), (2,1,'member'), (3,1,'member'),
(2,2,'admin'), (4,2,'member'), (5,2,'member'),
(3,3,'admin'), (1,3,'member'), (5,3,'member'),

```

Events

```

INSERT INTO Event (Title, OrganizingClubID, EventDate, Capacity)VALUES
('Hackathon 2025', 1, '2025-03-10', 5),
('Tech Quiz', 1, '2025-04-05', 3),
('Annual Drama Show', 2, '2025-03-20', 4),
('Robotics Workshop', 3, '2025-04-15', 3),
('Science Fair', 3, '2025-05-01', 10),
('Poetry Night', 2, '2025-06-01', 5);

```

Registrations

```
INSERT INTO Registration (StudentID, EventID, RegistrationDate, Attended)
VALUES
(1,1,'2025-03-01',TRUE),      (2,1,'2025-03-02',TRUE),
(3,1,'2025-03-03',TRUE),      (4,1,'2025-03-04',FALSE),
(5,1,'2025-03-05',TRUE),      (2,2,'2025-03-21',TRUE),
(1,2,'2025-03-20',TRUE),
(3,2,'2025-03-22',FALSE),      (2,3,'2025-03-10',TRUE),
(4,3,'2025-03-11',TRUE),      (5,3,'2025-03-12',TRUE), (6,3,'2025-03-13',FALSE),
3,4,'2025-04-10',TRUE),      (5,4,'2025-04-11',TRUE),
(1,5,'2025-04-20',FALSE);
```